

In Silico Evaluation and 25 ns Molecular Dynamics Simulation of Human Hsp70-Curcumin Complex

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Code & Reproducible Pipeline: <https://github.com/beshoyessam1027-arch>

Abstract

Heat Shock Protein 70 (Hsp70) is a crucial molecular chaperone overexpressed in various malignancies, rendering it an attractive therapeutic target in cancer computational biology. In this study, we established an integrated Computer-Aided Drug Design (CADD) and Molecular Dynamics (MD) simulation pipeline to evaluate the binding affinity, structural stability, and interaction kinetics of Curcumin against Human Hsp70 (PDB ID: 7GYI, resolution 1.90 Å). Molecular docking via AutoDock Vina yielded a strong binding affinity of -7.132 kcal/mol within the ATP-binding domain, mediated by key hydrogen bonds (Lys-1296 and Glu-269). Subsequent 25 ns MD simulations executed in GROMACS under physiological conditions (310 K, 1 bar) demonstrated high conformational stability. Trajectory analysis via R confirmed system convergence at ~ 15 ns with a stable Root Mean Square Deviation (RMSD) plateau of ~ 0.28 nm and a constant Radius of Gyration ($R_g \approx 2.21$ nm). Notably, a key ligand pose adjustment occurred at ~ 13 ns, optimizing steric fit and maintaining 1–3 persistent hydrogen bonds. ADME/Tox profiling via SwissADME and ADMETlab 3.0 confirmed favorable drug-likeness and low toxicity. This open-source workflow provides structural insights into Hsp70 inhibition for oncology therapeutics.

METHODS

Protein Preparation & Binding Site Mapping:

The 3D crystallographic structure of Human Hsp70 (PDB: 7GYI) was retrieved from the RCSB PDB. Water molecules and heteroatoms were removed, and missing hydrogen atoms were assigned at physiological pH. Active site residues surrounding the ATP-binding pocket were mapped using P2RANK and literature control data.

Ligand Docking & ADMET Profiling:

Curcumin was optimized, and gas-phase energy minimization was performed. Docking was executed using AutoDock Vina within a grid box centered on the active pocket. Pharmacokinetic parameters, drug-likeness (Lipinski’s Rule of Five), and safety profiles (hERG inhibition, hepatotoxicity) were predicted via SwissADME and

ADMETlab 3.0.

Molecular Dynamics (MD) Simulation Setup:

MD simulations were conducted using GROMACS in a Linux/WSL environment. Protein topology was generated using the CHARMM36 force field, while ligand parameters (`curcumin.itp`) were derived via CGenFF. The complex was solvated in a cubic box with TIP3P water molecules and neutralized with 0.15 M Na^+/Cl^- ions. Energy minimization was performed using the Steepest Descent algorithm (< 1000 kJ/mol/nm). System equilibration was carried out under NVT (310 K, 100 ps) and NPT (1 bar, 100 ps) ensembles. A production run of 25 ns (12,500,000 steps) was executed.

Trajectory Analysis & Visualization:

GROMACS tools (`gmx rms`, `gmx rmsf`, `gmx gyrate`, `gmx hbond`) were utilized to extract trajectory metrics. Custom R scripts with `ggplot2` were developed for statistical plotting, and PyMOL was used for molecular rendering and trajectory animations.

RESULTS & DISCUSSION

Binding Affinity & Interaction Profiling:

Curcumin bound efficiently into the N-terminal nucleotide-binding domain (NBD) of Hsp70, exhibiting a binding score of -7.132 kcal/mol. Protein-Ligand Interaction Profiler (PLIP) analysis revealed robust polar contacts anchored by Lys-1296 and Glu-269, alongside stabilizing hydrophobic interactions.

Structural Stability & Compactness (RMSD & R_g):

The backbone RMSD of the Hsp70-Curcumin complex stabilized rapidly after an initial equilibration phase, achieving a steady state at ~ 15 ns (mean RMSD ≈ 0.28 nm). Radius of Gyration (R_g) analysis demonstrated a consistent plateau at 2.20–2.22 nm, indicating that Curcumin binding preserves the overall structural compactness and structural integrity of the protein without causing unfolding.

Residue Flexibility & Hydrogen Bond Kinetics:

RMSF analysis revealed low atomic fluctuations ($<$

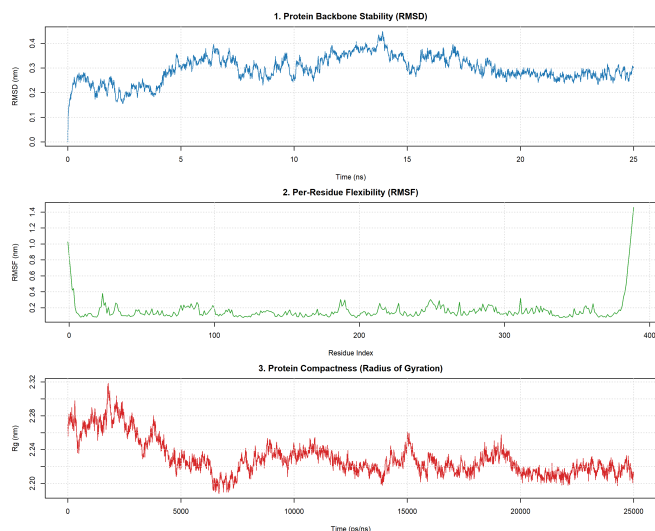


Figure 1: Molecular dynamics trajectory analysis of Hsp70 backbone over 25 ns. (1) Backbone RMSD showing system convergence at ~ 15 ns. (2) Residue-level RMSF indicating localized rigidity in the binding pocket (< 0.2 nm). (3) Radius of Gyration (R_g) confirming preserved overall protein compactness (~ 2.21 nm).

0.2 nm) across active site residues, confirming localized structural rigidity upon ligand engagement. Interestingly, a conformational transition was recorded at ~ 13 ns, where Curcumin underwent a subtle pose adjustment within the ATP pocket. This reorientation strengthened the hydrogen bonding network, increasing persistent H-bonds from 1–2 transient interactions to 2–3 continuous bonds for the remainder of the 25 ns run.

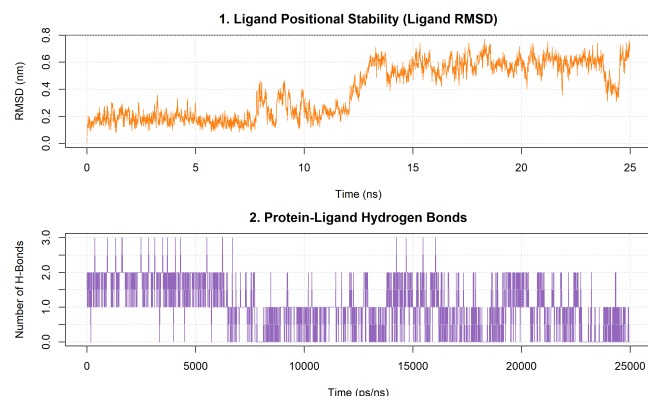


Figure 2: Ligand stability and hydrogen bonding kinetics of Curcumin bound to Hsp70. (1) Ligand RMSD trajectory displaying a structural reorientation at ~ 13 ns. (2) Time-dependent protein-ligand hydrogen bond profiles throughout the 25 ns simulation.

CONCLUSION

This study highlights Curcumin as a novel, highly stable inhibitor of Human Hsp70. The 25 ns MD trajectory confirms that Curcumin forms a resilient complex that maintains the structural compactness of Hsp70. The computational workflow, analytical R code, and MD trajectory assets are publicly available for further experimental validation.

DATA & CODE AVAILABILITY

Raw simulation parameters, topology files, R analysis scripts, and high-resolution animations are open-sourced at GitHub:

<https://github.com/beshoyessam1027-arch>